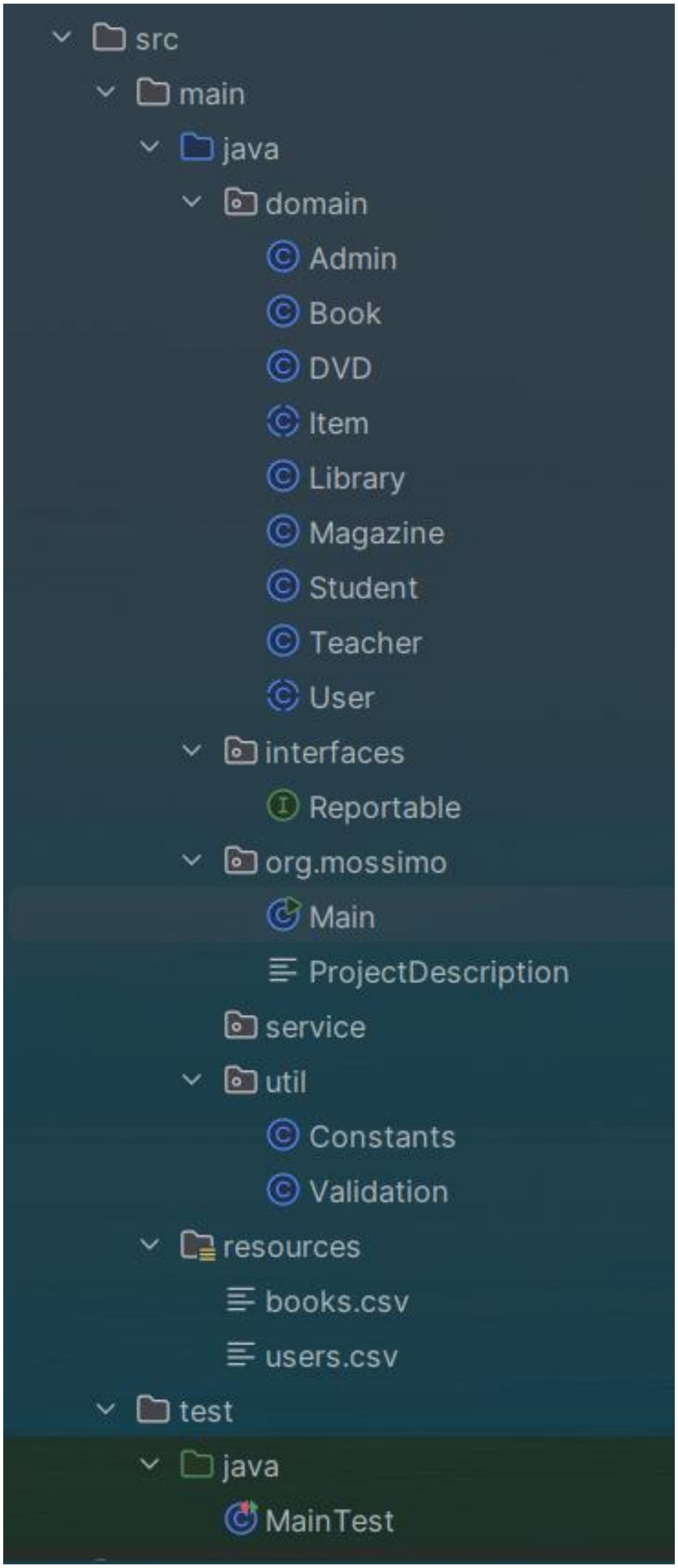
Library Management System — User Guide

This guide explains how to use the Library Management System inside IntelliJ IDEA. Follow each section step-by-step and insert screenshots where indicated.

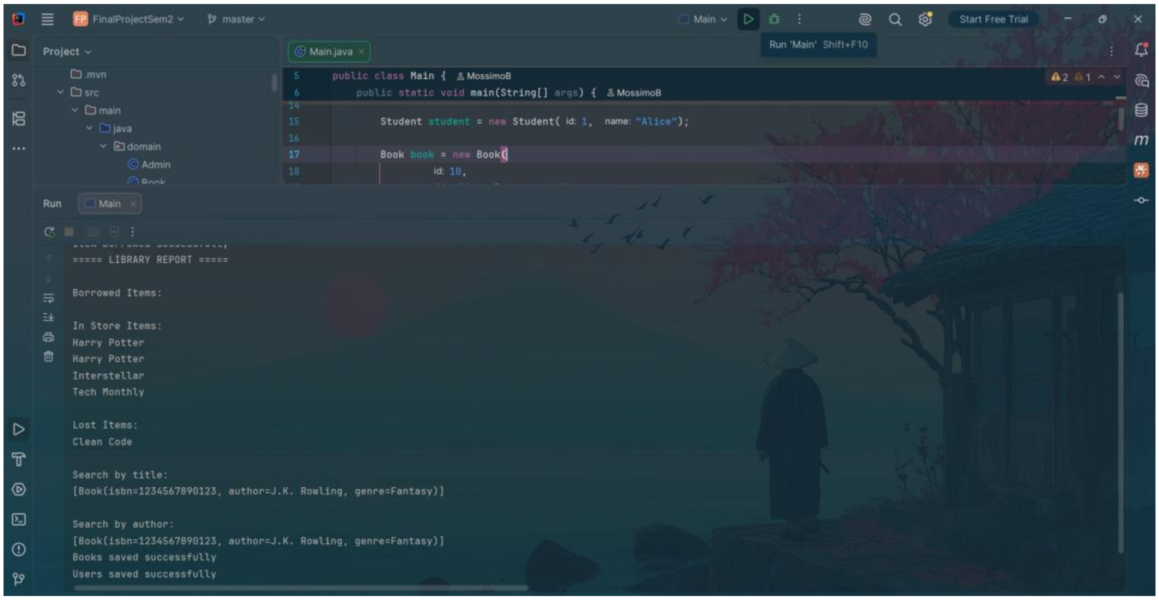
1. Opening the Project in IntelliJ IDEA

- Open IntelliJ IDEA.
- Click 'Open' on the main menu.
- Select the LibraryManagementSystem project folder.
- Wait for Maven dependencies to load.
- Make sure the src/main/java and src/test/java folders appear in the Project panel on the left.


src
main
java
domain
Admin
Book
DVD
Item
Library
Magazine
Student
Teacher
User
interfaces
Reportable
org.mossimo
Main
ProjectDescription
service
util
Constants
Validation
resources
books.csv
users.csv
test
java
MainTest

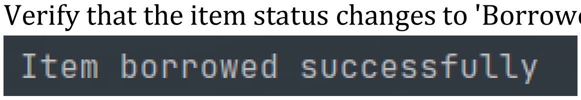
2. Running Main.java

- In the Project panel, navigate to src/main/java.
- Open the Main.java file.
- Click the green Run button beside the main() method.
- The console output will appear at the bottom of IntelliJ.
- Verify that books and users are loaded successfully.



3. Borrowing an Item

- Inside Main.java, locate the borrowItem() method call.
- Run the program again using the green Run button.
- Check the console output for 'Item borrowed successfully'.
- Verify that the item status changes to 'Borrowed'.



4. Returning an Item

- Inside Main.java, locate the returnItem() method call.
- Run the program again.
- Check that the item status changes back to 'In store'.
- Verify the returned item is pushed into the Stack collection.

5. Recursive Search

- Locate the recursiveSearch() method call inside Main.java.
- Run the program.
- Check the console output for the matching item title.
- Verify that the recursive search successfully finds the item.

6. Stream Search

- Locate the streamSearch() method call inside Main.java.
- Run the program.
- Verify that duplicate book copies only appear once in the output.
- Check the console for the search results.

```
System.out.println(library.streamSearch( title: "Harry Potter"));
```

7. Search by Author

- Locate the searchByAuthor() method call inside Main.java.
- Run the program.
- Check that books written by the specified author are displayed.
- Verify that the search is case-insensitive.

```
System.out.println(library.searchByAuthor("J.K. Rowling"));
```

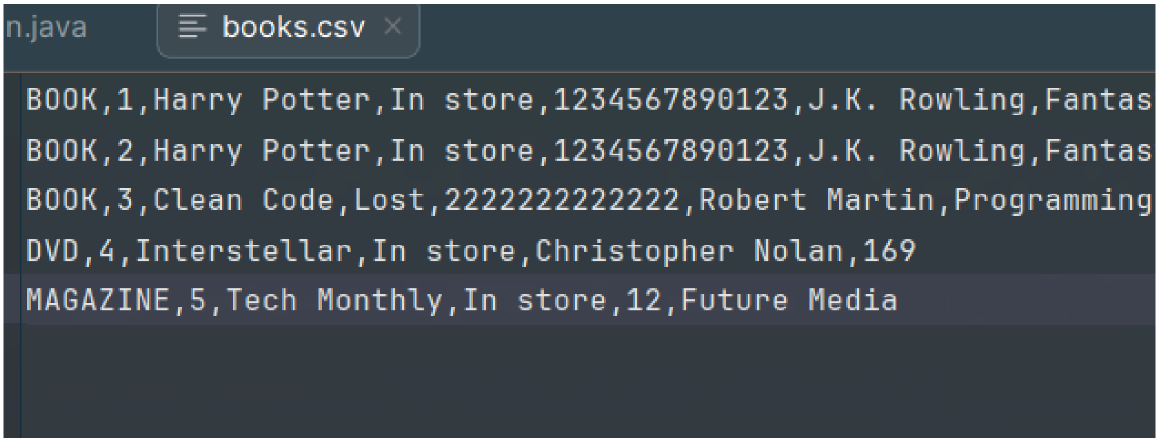
8. Admin Report

- Locate the generateReport() method call inside Main.java.
- Run the program.
- Verify that Borrowed, In Store, and Lost items appear in the console.
- Check that the Admin object generates the report correctly.

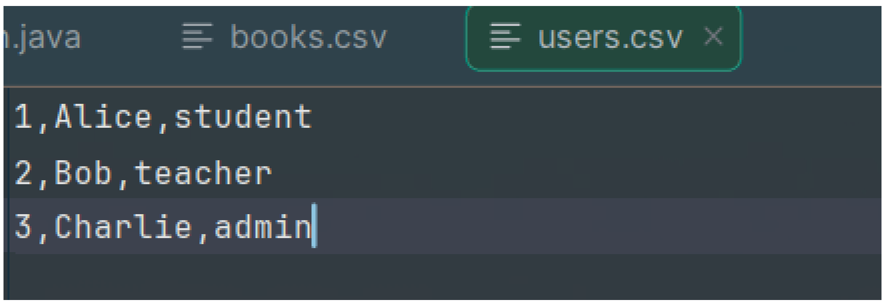
```
admin.generateReport();
```

9. CSV Files

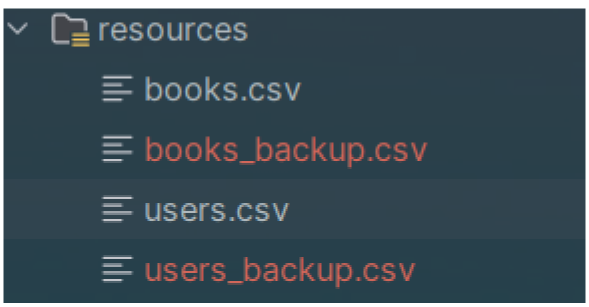
- In the Project panel, open src/main/resources.
- Open books.csv and users.csv.
- Run Main.java to generate backup files.
- Refresh the resources folder if necessary.
- Verify that books_backup.csv and users_backup.csv are created.




```
BOOK,1,Harry Potter,In store,1234567890123,J.K. Rowling,Fantas
BOOK,2,Harry Potter,In store,1234567890123,J.K. Rowling,Fantas
BOOK,3,Clean Code,Lost,2222222222222,Robert Martin,Programming
DVD,4,Interstellar,In store,Christopher Nolan,169
MAGAZINE,5,Tech Monthly,In store,12,Future Media
```

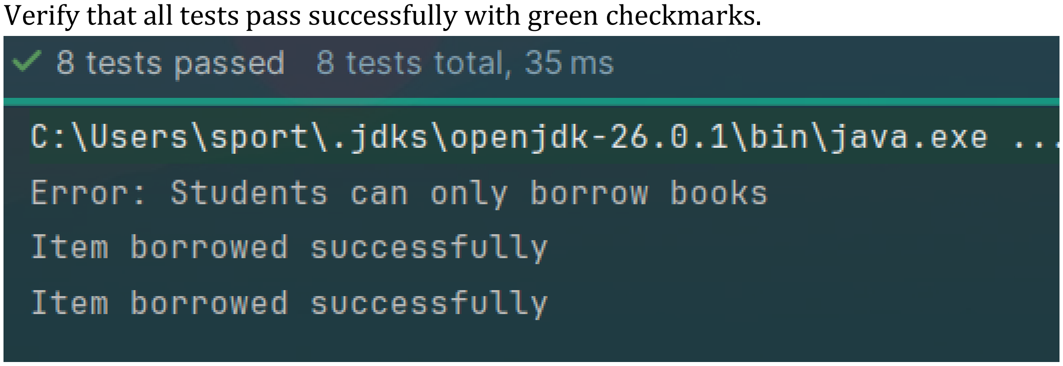



```
1,Alice,student
2,Bob,teacher
3,Charlie,admin
```



10. Running Unit Tests

- Open MainTest.java inside src/test/java.
- Click the green Run button beside the test class or beside individual test methods.
- Wait for IntelliJ to execute all JUnit tests.
- Verify that all tests pass successfully with green checkmarks.




```
C:\Users\sport\.jdks\openjdk-26.0.1\bin\java.exe ...
Error: Students can only borrow books
Item borrowed successfully
Item borrowed successfully
```

11. UML Class Diagram

- Open Visual Paradigm or diagrams.net.
- Create the UML diagram using the project classes and relationships.
- Export the final diagram as a PNG image.
- Insert the image into this section.